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ईएनटी शल्य चिकित्सा उपकरण — पैटर्सन प्रकार पैटर्न बायोप्सी संदंश — विशिष्टि (पहला पुनरीक्षण)

Ent Surgery Instruments — Paterson's Type Pattern Biopsy Forceps— Specification (First Revision)

11.040.30

MHD 4- Ear, Nose, Throat and Head & Neck Surgery (ENT - H&N)
Instruments

Last date for Comment:
XXXXXXX

FOREWORD

(Formal clause will be added later)

This standard was originally published in 1977 and Amendment No. 1 to the standard was published in 1984.

This first revision of the Standard has been undertaken to incorporate the following changes:

- a) Alignment with updated normative references, adoption of the latest format of Indian Standards.
- b) Amendment No. 1 published in 1984 is incorporated in the standard.
- b) The title has been changed from “Specification for Forceps, Biopsy. Paterson's pattern's Pattern” to “Ent Surgery Instruments — Paterson's Type Pattern Biopsy Forceps — Specification” as per latest format.
- c) The clause on shape and dimensions has been made recommendatory rather than mandatory, to allow flexibility on shape and dimensions by mutual agreement between the manufacturer and the purchaser.

For the purpose of deciding whether a particular requirement of this standard is complied with the final value, observed or calculated, expressing the result of a test or analysis shall be rounded off in accordance with IS 2: 2022 ‘Rules for rounding off numerical values (*second revision*)’. The number of significant places retained in the rounded-off value should be same as that of the specified value in this standard.

1 SCOPE

This standard covers dimensional and other requirements for Paterson's type pattern biopsy forceps of sizes 20, 25, 30, 35, 40, 45, 50, 55 and 60 cm used in ENT surgery.

2 REFERENCES

The standards given below contain provisions which, through reference in this text, constitute provisions of this standard. At the time of publication, the editions indicated were valid. All standards are subject to revision, and parties to agreements based on this standard are encouraged to investigate the possibility of applying the most recent editions of these standards.

<i>IS No.</i>	<i>IS Title</i>
IS 1501 : Part 1 : 2025	Metallic materials Vickers hardness test Part 1: Test method (Sixth Revision)
IS 3642 : Part 1 : 1990	Surgical instruments - Specification: Part 1 non - Cutting, articulated instruments (Second Revision)
IS 6603 : 2024	Stainless Steel Semi-Finished Products, Bars, Wire Rods and Bright Bars " Specification (Second Revision)
IS 7531 : 1990	Surgical instruments - Corrosion resistance of stainless steel surgical instruments - Methods of tests (First Revision)

3 SHAPE AND DIMENSIONS

3.1 The shape and dimensions should be as shown in Fig. 1 or as agreed between manufacturer and user.

3.2 A deviation of ± 2.5 percent may be allowed on all dimensions.

3.3

A deviation of $\pm 2^\circ$ may be allowed on all angular dimensions.

4 MATERIAL

The forceps shall be made of stainless steel conforming to Designation X40Cr13 of [IS 6603](#) The materials shall be stable to conditions employed for sterilization of surgical tools such as steam sterilization.

5 WORKMANSHIP AND FINISH

5.1

The forceps shall be free from scales, burrs, cracks, pits, tool marks and other surface defects.

5.2

All edges shall be smoothly rounded off except the cutting edge, which should be as shown in **Fig.1**.



Figure 1 :

E =100, L = 200, 250, 300, 350, 400, 450, 500, 550 and 600. *See [IS 3642 : Part 1](#)

FIG. 1 Paterson's Type Pattern Biopsy Forceps

5.3

The jaws shall open and close fully and freely without any stiffness.

5.4 The forceps shall be passivated and shall have matt finish.

6 HEAT TREATMENT AND HARDNESS

6.1

General

The trocar portion shall be heat treated using hardening and tempering procedures to produce specified hardness.

6.2

Hardness

The Vickers hardness of the finished instrument shall be within the range of 540 to 600 HV, when tested in accordance with [IS 1501 : Part 1](#)

7 TEST

7.1

Performance

The forceps shall be made to cut a 0.5 mm thick card-board sheet. The forceps shall do this easily and cleanly. This shall be repeated 10 times. During or on completion of the test, the cutting edges of the forceps shall show no sign of damage.

7.2

Flexibility Test — The flexibility shall be tested in the following manner:

7.2.1 The jaws shall be made to bite a piece of firm rubber about 3.00 mm thick and the finger loops closed as fully as possible. On release, the forceps shall not show any sign of damage or the upper jaw shall not become loose at the hinges.

7.2.2 A force of moderate degree shall be applied by the fingers to the shank attempting to bend the shank on the width. This shall be repeated at several points along the shank. The test shall, then, be repeated in a plane at right angles to the first. On completion of the test the shank shall not have acquired permanent set.

7.3

Corrosion resistance

The forceps shall be tested in accordance with [IS 7531](#). There shall not be any signs of corrosion and the arms shall move freely at the joint at the end of the test.

8 MARKING

8.1

The instrument shall be legibly and indelibly marked with the following information:

- a) Manufacturer's name, initials or recognized trade-mark,
- b) The letters 'SS' or the words 'Stainless Steel', and
- c) The country of manufacture.

8.2

BIS Certification Marking

The product(s) conforming to the requirements of this standard may be certified as per the conformity assessment schemes under the provisions of the *Bureau of Indian Standards Act, 2016* and the Rules and Regulations framed thereunder, and the product(s) may be marked with the Standard Mark.

9 PACKING

The forceps shall be wrapped in moisture-proof paper or packed in polyethylene bag. Each forceps shall then be packed in individual cartons. The forceps may also be packed as agreed to between the purchaser and the supplier.